

SUPPLY MANAGEMENT SYSTEM

Task Addendum

Supplier Reporting Suite and Supplier Scorecard

Maps to FSD Addendum 23 | ASP.NET Core 8 | EF Core 8 | Hangfire
Version 1.0 | 2026

FSD source	FSD Addendum 23 - Supplier Reporting Suite and Supplier Scorecard
Total tasks	8
Total estimated dev days	21 days (~5 working weeks)
Critical priority tasks	2
Total unit test cases	46
Epics	4 epics: supplier reports, scoring engine, dashboard UI, gap reports

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1. FSD Traceability Summary

Every task maps to FSD Addendum 23.

Task ID	Title	Days	FSD Ref	Priority
SC-001	Supplier-Wise Ledger Summary report	2d	Sec 2	High
SC-002	Supplier Orders Report with delivery performance variance	3d	Sec 3	High
SC-003	Scorecard schema: SupplierScoreSnapshots, ScorecardDimensionWeights, GrnScoreDetails	3d	Sec 4, 5.3, 5.4	Critical
SC-004	Per-GRN scoring engine (ScoreGrnAsync) with all 5 dimensions	3d	Sec 4.1, 6	Critical
SC-005	Supplier aggregate recalculation and snapshot storage	2d	Sec 6.2	High
SC-006	Supplier Scorecard Dashboard - React UI	3d	Sec 5.2	High
SC-007	Grade F supplier warning on PO creation and Supplier Master	2d	Sec 5.2	High
SC-008	Supplier Comparison, GRN Quality, Spend Analysis, and Delivery Heatmap reports	3d	Sec 7	Medium
TOTAL	8 tasks	21d		

2. Detailed Task Register

Epic 1 - Supplier Reports: Ledger Summary and Orders (FSD Sections 2, 3)

Cross-supplier ledger summary with drill-down and the comprehensive orders-by-supplier report with delivery performance analysis.

SC-001	High	2d	Sec 2	Supplier-Wise Ledger Summary report
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Build the cross-supplier ledger summary showing one row per supplier with total invoiced, total paid, outstanding balance, and drill-down to the per-supplier ledger from Addendum 22.

Development tasks	Unit test cases
- GET /api/reports/supplier-ledger-summary - aggregates SupplierLedgerEntries per supplier: total debits (invoiced), total credits (paid), outstanding balance, advance balance, last transaction date - Filters: date range, supplier category, supplier status, minimum outstanding threshold - Sort: by outstanding_balance descending (default), supplier_name, last_transaction_date - Drill-down link per row to GET /api/suppliers/{id}/ledger (Addendum 22) with the report's date range as query params - PDF and Excel export via QuestPDF / ClosedXML	- Summary row for a supplier with 3 invoices (total 150,000) and 2 payments (total 80,000) shows outstanding 70,000 - Date range filter correctly excludes ledger entries outside the range - Minimum outstanding threshold filter excludes suppliers below the threshold - Drill-down link correctly passes the date range to the per-supplier ledger endpoint - PDF export produces a correctly formatted file with all visible rows

Acceptance criteria	- Cross-supplier view with accurate aggregation - Drill-down seamlessly opens the existing per-supplier ledger - Export functional in both PDF and Excel
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SC-002	High	3d	Sec 3	Supplier Orders Report with delivery performance variance
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Build the orders-by-supplier report showing PO summary per supplier and per-PO delivery performance (actual vs expected), colour-coded for on-time/late.

Development tasks	Unit test cases
- GET /api/reports/supplier-orders - summary row per supplier: total_po_count, total_po_value, avg_po_value, fully_received/partially_received/pending/cancelled PO counts - Detail drill-down per supplier: every PO with po_number, date, total, status, qty_ordered vs qty_received, grn_count, delivery_variance_days - Delivery variance = first GRN received_at minus PO expected_delivery_date, in days (negative = early, zero = on time, positive = late)	- Supplier with 5 POs shows correct counts for each status (fully received, partial, pending, cancelled) - PO with expected_delivery_date June 10 and first GRN received_at June 15 shows variance = +5, coded amber - PO with no expected_delivery_date shows neutral variance (no colour) - Filtering by delivery performance = late returns only POs with positive variance - Grand total correctly sums across all filtered suppliers

• Colour coding: green (on time or early), amber (1-7 days late), red (8+ days late). Colour sent as a response field for React rendering • Filters: date range, supplier, PO status, delivery performance category (on-time/late/all) • Grand total summary row across all suppliers	
Acceptance criteria	• Delivery variance calculated accurately from actual GRN vs PO dates • Colour coding correctly applied at the threshold boundaries • Drill-down from summary to individual POs works without page reload

Epic 2 - Scoring Engine and Schema (FSD Sections 4, 5, 6)

The automated scoring engine that computes per-GRN dimension scores, aggregates into supplier snapshots, and stores them for the dashboard.

SC-003	Critical	3d	Sec 4, 5.3, 5.4	Scorecard schema: SupplierScoreSnapshots, ScorecardDimensionWeights, GrnScoreDetails
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Create all scorecard entities in suppliers_schema, seed the 5 default dimensions with their default weights, and build the weight configuration API.

Development tasks	Unit test cases
- Add SupplierScoreSnapshots entity (15 fields per FSD Section 5.3) to SuppliersDbContext - Add ScorecardDimensionWeights entity (7 fields per FSD Section 5.4) to SuppliersDbContext - Add GrnScoreDetails entity (per-GRN raw dimension scores): grn_score_id, grn_id FK, supplier_id FK, delivery_points, quantity_points, quality_points, price_points, documentation_points, total_raw_score, weighted_score, scored_at - EF Core migration: dotnet ef migrations add AddSupplierScorecard - Seed 5 default dimensions: DELIVERY (25%, max 25), QUANTITY (25%, max 25), QUALITY (25%, max 25), PRICE (15%, max 15), DOCUMENTATION (10%, max 10) - GET /api/admin/scorecard-weights - returns current weights - PUT /api/admin/scorecard-weights - validates weights sum to 100, updates all dimensions atomically. Admin role only.	- After migration, 5 dimension rows exist with correct default max_points and weight_percentage summing to 100 - PUT weights with values summing to 99 returns 400 validation error - PUT weights with values summing to 100 succeeds and persists - Non-Admin role calling PUT returns 403 - GET weights returns all 5 dimensions with current values

Acceptance criteria	- Schema correctly created with all entities - Default weights correctly seeded - Weight validation enforces sum = 100
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SC-004	Critical	3d	Sec 4.1, 6	Per-GRN scoring engine (ScoreGrnAsync) with all 5 dimensions
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Implement ISupplierScoringService.ScoreGrnAsync that calculates raw points per dimension for a single GRN using the rules from FSD Section 4.1, and the Hangfire wiring to call it on every GRN approval.

Development tasks	Unit test cases
- Implement ScoreGrnAsync(grnId): loads GRN, linked PO, PO lines, GRN lines, inspection results (Addendum 19), and invoice data - Delivery Timeliness: compare GRN.received_at vs PO.expected_delivery_date using the 5-tier point scale (25/20/15/10/5). Default 20 if no expected date.	- GRN delivered on time with 100% quantity match, all items passing inspection, exact price match, full documentation: scores 25+25+25+15+10 = 100 raw, 100 weighted - GRN 10 days late with 85% quantity, 75% inspection pass, 8% price variance, one missing doc: scores 10+10+10+6+4 = 40 raw

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| <ul style="list-style-type: none"> Quantity Accuracy: compare cumulative GRN qty_accepted across all GRNs for the PO line vs PO qty_ordered. 5-tier scale (25/20/15/10/5) based on percentage match. Quality/Inspection: if GRN.requires_inspection = true, percentage of lines with inspection result = Pass. 5-tier scale. Default 20 if no inspection. Price Compliance: compare GRN/Invoice unit prices vs original PO unit prices per line. 5-tier scale (15/12/9/6/3) based on variance percentage. Documentation Completeness: check invoice received within payment terms, delivery note attached, certificates present. 3-tier scale (10/7/4/2). Calculate weighted_score = sum of (dimension_points / dimension_max * dimension_weight_percentage) across all active dimensions Write GrnScoreDetails row with all dimension points, total_raw_score, and weighted_score Subscribe to GrnApprovedEvent (MediatR): enqueue BackgroundJob.Enqueue() => scoringService.ScoreGrnAsync(grnId)) - non-blocking, same pattern as timeline writes | <ul style="list-style-type: none"> GRN with requires_inspection = false gets default quality score of 20 GRN with no expected_delivery_date on PO gets default delivery score of 20 ScoreGrnAsync failure does not roll back the GRN approval Weighted score correctly adjusts when Admin changes dimension weights (verified by rescoring the same GRN after a weight change) |
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Acceptance criteria

- All 5 dimensions correctly scored from live GRN/PO/Invoice/Inspection data
- Scoring is non-blocking and runs via Hangfire
- Default handling correct for missing data (no expected date, no inspection)

SC-005	High	2d	Sec 6.2	Supplier aggregate recalculation and snapshot storage
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Implement RecalculateAllAsync and RecalculateSupplierAsync that average per-GRN weighted scores into a SupplierScoreSnapshot with grade and trend, on a configurable Hangfire schedule.

Development tasks	Unit test cases
- RecalculateSupplierAsync(supplierId, period): query all GrnScoreDetails for this supplier within the period, compute average weighted_score, determine grade from the A/B/C/D/F scale (FSD Section 5.1), compute trend by comparing to the previous period's snapshot - RecalculateAllAsync(period): iterate all active suppliers with at least one scored GRN in the period, call RecalculateSupplierAsync for each - Trend logic: current composite_score vs previous snapshot composite_score. Difference > +5 = IMPROVING, -5 to +5 = STABLE, < -5 = DECLINING - Register Hangfire RecurringJob: configurable frequency (daily/weekly/monthly) via appsettings SupplierScorecard:RecalculationFrequency - POST /api/supplier-scorecard/recalculate - Admin-triggered full recalculation	- Supplier with 4 GRNs scoring 90, 85, 80, 95: composite = 87.5, grade = B - Supplier with previous period composite 70, current 82: trend = IMPROVING - Supplier with previous period composite 85, current 83: trend = STABLE - Supplier with zero GRNs in the period gets no snapshot (not scored, excluded from dashboard) - Admin-triggered recalculation overwrites the existing snapshot for the same period - Hangfire job correctly runs at the configured frequency

• POST /api/supplier-scorecard/{supplierId}/recalculate - Admin-triggered single supplier	
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Acceptance criteria	• Aggregate correctly averages all per-GRN scores in the period • Grade derived accurately from the composite score thresholds • Trend correctly compares current vs previous period
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Epic 3 - Scorecard Dashboard and Grade F Warning (FSD Section 5.2)

The React scorecard dashboard with ranking table, detail drilldown, summary cards, and the Grade F warning on PO creation.

SC-006	High	3d	Sec 5.2	Supplier Scorecard Dashboard - React UI
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Build the scorecard dashboard screen with the supplier ranking table, per-dimension detail panel with charts, summary cards, and evaluation period selector.

Development tasks	Unit test cases
- React: new screen at Suppliers > Supplier Scorecard. GET /api/supplier-scorecard?periodStart={}&periodEnd={} returns ranked supplier list from SupplierScoreSnapshots - Ranking table columns: Rank, Supplier Name, Grade (letter badge with colour per Section 5.1), Composite Score, Delivery / Quality / Quantity / Price / Documentation individual scores, GRN Count, Trend arrow (up green / flat grey / down red) - Evaluation period selector: preset buttons (3m / 6m / 12m) and custom date range picker - Dashboard summary cards at top: total suppliers scored, average composite score, grade distribution (A:5, B:12, C:8, D:3, F:1), most improved supplier, most declined supplier - Clicking a supplier row opens a slide-over detail panel - GET /api/supplier-scorecard/{supplierId} returns per-dimension breakdown, per-GRN score list, and historical trend data - Detail panel: radar/spider chart showing 5 dimensions, scrollable list of individual GRN scores with drill-down link to GRN detail, line chart showing composite score trend over last 4 periods	- Dashboard loads with correct ranking sorted by composite score descending - Grade badge colours match: A=green, B=light-green, C=amber, D=red, F=dark-red - Period selector changing from 3m to 12m refetches and updates all data - Summary cards show correct grade distribution matching the visible table rows - Detail panel radar chart correctly renders all 5 dimensions with proportional values - Trend arrow correctly shows up/flat/down based on snapshot trend field

Acceptance criteria	- Dashboard renders accurate data from snapshots without recalculating on every load - Detail panel provides actionable per-GRN drill-down - Period selection dynamically updates all components
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SC-007	High	2d	Sec 5.2	Grade F supplier warning on PO creation and Supplier Master
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Add visual alert for Grade F suppliers: a warning banner on the Supplier Master detail screen and a non-blocking warning on the PO creation screen when a Grade F supplier is selected.

Development tasks	Unit test cases
React: on the Supplier Master detail screen, if the supplier's latest snapshot grade = F, render a red alert banner: 'This supplier is rated F'	- Selecting a Grade F supplier on PO creation shows the red warning message - Selecting a Grade A supplier shows no warning

(Disqualification Candidate). Review the Scorecard before placing new orders.' • React: on PO creation, when a supplier is selected via the supplier dropdown, check the supplier's latest grade from a lightweight GET /api/suppliers/{id}/score-summary (returns grade and composite_score only) • If grade = F: show a non-blocking amber/red warning below the supplier field: 'Warning: {supplier_name} is rated F. Procurement Manager approval is recommended before proceeding.' PO creation is NOT blocked - this is advisory only. • If grade = D: show a softer amber warning: '{supplier_name} is rated D (Underperforming). Consider alternative suppliers.' • Grades A/B/C show no warning. The supplier dropdown is never filtered by grade - all active suppliers remain selectable. • Procurement Manager can acknowledge and dismiss the warning to proceed	• Warning is dismissible and does not block PO save • Supplier Master detail screen for a Grade F supplier shows the alert banner • Supplier with no score snapshot (never evaluated) shows no warning • Grade D supplier shows the softer amber warning, not the red F warning
Acceptance criteria	• Warnings are advisory only - never block PO creation • Warning text correctly references the supplier name and grade • No warning shown for unscored or A/B/C suppliers

Epic 4 - Additional Missing Reports (FSD Section 7)

Supplier comparison, GRN quality analysis, spend analysis, and delivery performance heatmap.

SC-008	Medium	3d	Sec 7	Supplier Comparison, GRN Quality, Spend Analysis, and Delivery Heatmap reports
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Implement the 4 remaining reports identified as gaps: side-by-side supplier comparison, GRN quality trends, spend concentration analysis, and the delivery performance calendar heatmap.

Development tasks	Unit test cases
- GET /api/reports/supplier-comparison?ids=a,b,c - accepts 2-5 supplier IDs; returns side-by-side columns: scorecard grade, composite score, order volumes, average delivery variance, rejection rate (from GRN inspection data), payment history summary (from Addendum 22 ledger) - GET /api/reports/grn-quality-analysis - inspection pass/fail/partial rates by supplier over time. Response includes per-supplier trend data (monthly pass rates) for chart rendering. Drill-down to specific failed GRN lines with rejection reasons. - GET /api/reports/supplier-spend-analysis - total invoiced amount per supplier over date range, broken down by product category/sub-category. Includes top-10 suppliers by value and spend concentration percentage (top 5 suppliers as % of total spend). - GET /api/reports/delivery-performance-heatmap?supplierId={id}&year={year} - returns 12 months of daily GRN data: date, on-time/late status, variance days. React renders as a GitHub-style contribution heatmap (green = on time, amber = 1-7 late, red = 8+ late, grey = no GRN). - All 4 endpoints use read-only ReportsDbContext	- Supplier comparison with 3 suppliers returns correct side-by-side data with no cross-contamination between columns - Comparison with more than 5 supplier IDs returns 400 - GRN quality analysis correctly calculates monthly pass rates from GRNLine inspection results - Spend analysis top-10 list is correctly sorted by invoiced amount descending - Spend concentration percentage correctly calculates top-5 suppliers as % of total - Heatmap for a supplier with zero GRNs in a month shows all grey for that month - Heatmap correctly colour-codes a GRN received 10 days late as red

Acceptance criteria	- All 4 reports return accurate data sourced from live transaction tables - Comparison limited to 2-5 suppliers as specified - Heatmap data structure is chart-ready for React rendering
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3. Sequencing Notes

The scoring engine (Epic 2) is the critical path. Reports and dashboard depend on scored data existing.

Dependency	Reason
SC-001, SC-002 independent	Supplier reports read from existing Addendum 22 ledger and PO/GRN data. Can start immediately.
SC-003 before SC-004	Scoring engine needs schema and dimension weights to exist.
SC-004 before SC-005	Aggregate recalculation averages per-GRN scores created by SC-004.
SC-005 before SC-006	Dashboard reads from SupplierScoreSnapshots written by SC-005.
SC-005 before SC-007	Grade F warning reads the supplier's latest snapshot grade.
SC-008 independent of SC-003-007	Gap reports read from existing PO/GRN/Invoice data. Can be parallelised.

SC-001, SC-002, and SC-008 can all be developed in parallel with the scoring engine (SC-003 to SC-007) since they read from pre-existing data, not scorecard entities.

- End of Document -